OFFERS ANALYTICS · PORTFOLIO PROJECT

Who should get the coupon?

Redemption propensity and uplift modelling on 2,469 households and 1.47M grocery transactions

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Two questions, not one

PROPENSITY

Who will redeem if we send the offer?

A ranking of targeted customers. Easy to measure, but it rewards customers who would have bought anyway.

UPLIFT

Who buys more because of the offer?

The incremental effect. This is what an offer budget should be spent on.

2,469
households, 53 weeks

1.47M
transactions

3
large campaigns, ~1,100 targeted each

8, 13 → 18
train on two, test on the next

Data: Dunnhumby “The Complete Journey” (completejourney_py). Features use only data from the 16 weeks before each campaign.

The retailer targeted its best customers

Before campaign 18 began, targeted households already shopped far more than the ones left out, so a simple targeted-vs-not comparison flatters the coupon.

2.9×

Pre-period spend

$898 targeted vs $305 not

2.7×

Shopping trips

30.5 vs 11.1 in 16 weeks

3×

More recent

6 vs 19 days since last trip

1.1 vs 0.1

Past redemptions

per household

Campaign 18: 1,129 targeted vs 1,215 not targeted, among 2,344 active households. Means from the 16 weeks before launch.

Five segments, very uneven targeting

Segment (K-means, k = 5)	Households	% targeted	Redeemed	Control vs treated
Loyal big-basket / coupon power users	266	91%	55%	25 control: too few
Frequent fuel & grocery shoppers	504	63%	12%	187 vs 317: usable
Rising spenders	857	53%	7.5%	402 vs 455: usable
Lapsing, low engagement	399	15%	10%	60 treated: too few
Light, occasional shoppers	318	18%	3.6%	56 treated: too few

Loyal customers redeem most, but almost all of them were mailed, so there is no comparison group to show whether the coupon changed their behaviour.

Campaign 18. “Redeemed” = share of targeted households that redeemed. Clustered on behaviour only (RFM, discount use, basket mix).

Coupon history predicts redemption best

Campaign 18 test set, targeted households

Metric	Calibrated XGBoost	Rule: past redemptions
ROC-AUC	0.84	0.78
PR-AUC (base rate 0.19)	0.63	0.56
Top-decile lift	3.7×	3.7× (tie)
Redeemers reached at 40% contact	81.8%	72.9%

The model ties the simple rule for the very top customers and pulls ahead as the mailing widens. Probabilities are calibrated: 0.187 predicted vs 0.190 actual.

What drives redemption

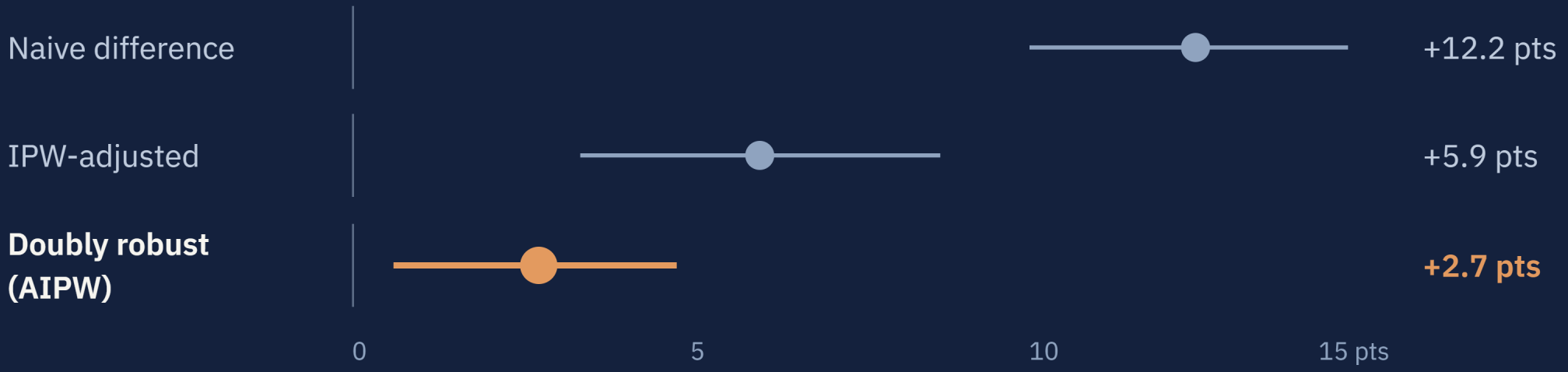
1. Past coupon use and coupon discounts
2. Share of spend on discounted items
3. Spend and basket size
4. Recency: lapsed shoppers redeem less

Dropping all 45 demographic columns left PR-AUC unchanged at 0.62: behaviour is enough.

Trained on campaigns 8 and 13 (grouped 5-fold CV by household), tested once on campaign 18. Model chosen on CV, not on the test set.

Most of the apparent effect was selection

Effect of being targeted on buying any couponed product during campaign 18, in percentage points, with 95% intervals.

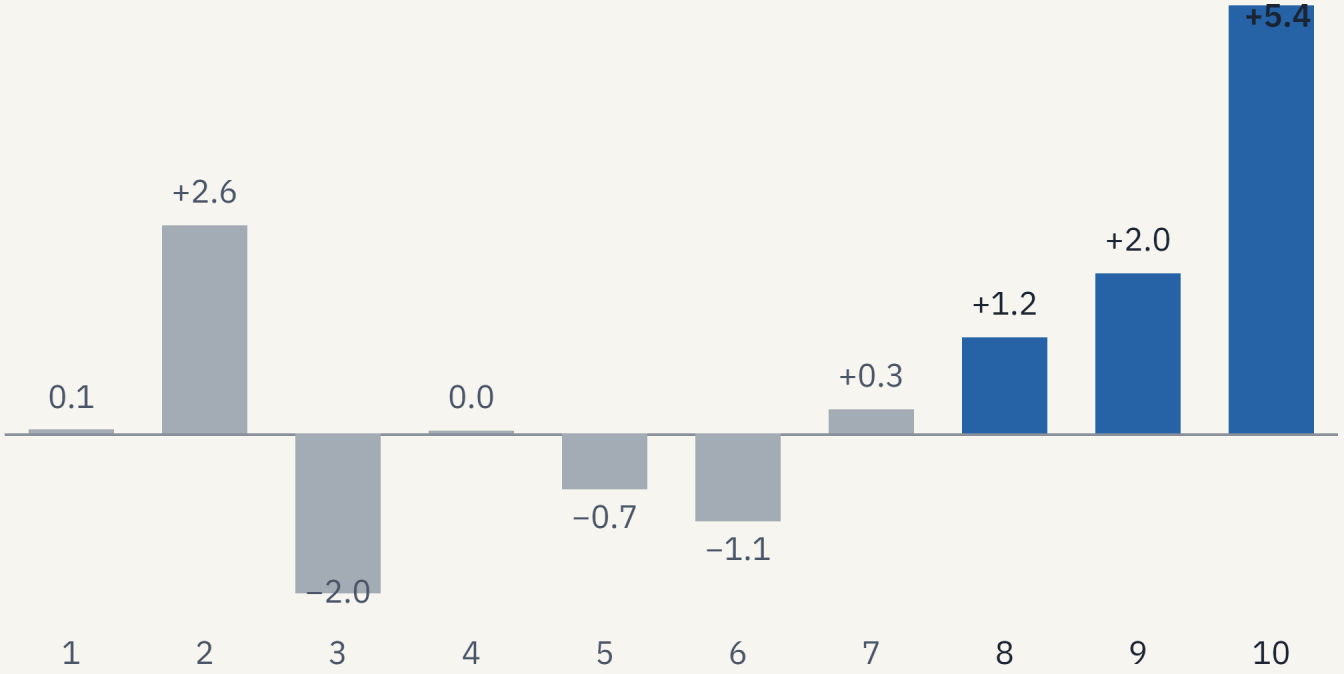


The coupon works, but about four-fifths of the naive effect came from who was targeted. On weekly spend the adjusted effect is about zero ($0.01, interval −$1.17 to +$1.10).

2,344 active households; households with extreme treatment propensity trimmed. Intervals: bootstrap 95%.

Uplift concentrates in a small group

Actual change in weekly coupon-product spend, treated minus control ($/week), by predicted-uplift decile



Decile of predicted uplift (10 = highest; blue = top three)

Top ~16%

Incremental spend peaks at +$1,474/week when the top 15.7% are treated. Mailing everyone ends below random.

Beats chance: Qini +532, permutation p < 0.002; bias-adjusted p = 0.014.

Caveat

Top deciles were 70–79% targeted vs 28–41% at the bottom (r = 0.85). Part of the signal may be the retailer's own selection.

T-learner (LightGBM), trained on campaigns 8 and 13, scored on campaign 18. Deciles are not strictly monotonic; Qini peak is cumulative across contacted households.

Recommendations

1

Hold out a random 10%

in every campaign. It is the only way to measure the offer's effect on the loyal customers who are almost always mailed.

2

Pilot uplift targeting

on the top ~15–20% by predicted uplift, against a randomly targeted arm of the same size.

3

Judge offers on incremental spend

not redemptions. Coupons lifted buying of couponed items by 2.7 pts but left weekly spend unchanged.

Limitations

Observational data only · 9 behavioural features stay imbalanced after weighting · weak overlap for Loyal, Lapsing and Light segments · one test campaign

Code and notebooks: github.com/azutshi1305-coder/coupon-uplift-model